

Module 9: Fundamental Concepts of Loops

Practice Task

Task 1 — Print "Hello" Five Times

Write a for loop that prints "Hello" 5 times.

```
for(let i = 1; i <= 5;i++){  
  console.log("Hello");  
}
```

Task 2 — Print Numbers 1 to 10

Write a for loop that prints numbers from 1 to 10.

```
for (let i = 1; i <= 10; i++) {  
  console.log(i)  
}
```

Task 3 — Countdown with While Loop

Write a while loop that prints numbers from 10 down to 1.

```
let count = 10;  
while( count >= 1){  
  console.log("Value of count" , count)  
  count--  
}
```

Task 4 — Repeat a Message

Using a while loop, print "Loop চলছে" exactly 7 times.

```
let initialMessage = 1;  
while (initialMessage <= 7) {  
  console.log("Loop " , initialMessage);  
  initialMessage++;  
}
```

Task 5 — Even Numbers

Write a for loop that prints all even numbers between 1 and 30.

```
for (let i = 2; i <= 30; i += 2) {  
  console.log("The number is ", i);  
}
```

Task 6 — Sum of Numbers

Write a for loop that calculates the sum of numbers from 1 to 20 and prints the result.

```
let sum = 0;  
for (let i = 1; i <= 20; i++) {  
  sum = sum + i;  
}  
console.log("The result is ", sum);
```

Task 7 — Multiplication Table

Ask the user for a number (or set a variable), then print its multiplication table from 1 to 10 using a for loop. Example: if the number is 7 → $7 \times 1 = 7$, $7 \times 2 = 14$, ... $7 \times 10 = 70$.

```
let number= 7;  
for(let i = 1; i <= 10; i++){  
  console.log(number , "X" , i , " = ", number*i,);  
}
```

Task 8 — Count Down from 20

Write a for loop that counts down from 20 to 1.

```
for (let count = 20; count >= 1; count--) {  
  console.log("Count is ", count);  
}
```

Task 9 — Sum of Even Numbers

Write a for loop that calculates the sum of all even numbers from 2 to 50.

```
let sumOfEven = 0;  
for (let i = 2; i <= 50; i+=2) {  
  sumOfEven = sumOfEven + i;  
}  
console.log("Sum of Even between 2 to 50 is", sumOfEven);
```

Task 10 — Using Break

Write a for loop from 1 to 50 that stops completely (break) as soon as it reaches a number greater than 30.

```
for (let i = 1; i <= 50; i++) {  
  if(i >= 30){  
    break;  
  }  
  console.log(i);  
}
```

Task 11 — Using Continue

Write a for loop from 1 to 20 that skips (continue) any number divisible by 4.

```
for (let i = 1; i <= 20; i++) {  
  if (i % 4 === 0) {  
    continue;  
  }  
  console.log(i);  
}
```

Task 12 — Do While Loop — Optional

Write a do while loop that prints numbers from 1 to 5. Then modify the condition so it is false from the start (for example, $i > 10$) and observe that the loop still runs once.

```
let index = 1;  
do {  
  console.log("index", index);  
  index++;  
} while (index <= 5);  
  
/* Modification */  
let ind = 1;  
do {  
  console.log("index", ind);  
  ind++;  
} while (index > 10);
```

Task 13 — Comparing Loop Types

Write the same task — printing numbers 1 to 5 — three times: once using for, once using while, and once using do while. Compare how the code structure differs for each.

```
// For Loop
for (let i = 1; i <=5; i++){
  console.log(i)
}

// While Loop
let number1 = 1;
while(number1 <= 5){
  console.log(number1);
  number1++
}

// Do-While Loop
let number2 = 1;
do{
  console.log(number2);
  number2++
}while(number2 <= 5);
```

Task 14 — Mini Challenge

Write a program that:

1. Loops from 1 to 100.
2. Skips (continue) any number divisible by 5.
3. Stops (break) completely once it reaches a number greater than 40.
4. Prints every number that passes both checks.

```
for(let i = 1; i <= 100 ; i++){
  if( i % 5 === 0){
    continue;
  }
  if(i > 40){
    break;
  }
  console.log(i);
}
```